

Microstructures and compressive properties of multicomponent AlCoCrFeNiMo_x alloys

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ABSTRACT

Multicomponent AlCoCrFeNiMo_x (x values in molar ratio, $x=0, 0.1, 0.2, 0.3, 0.4$ and 0.5) alloys were prepared using a well-developed copper mould casting. The effects of Mo element on the structure and properties of AlCoCrFeNi alloy were investigated by X-ray diffraction (XRD), scanning electron microscopy (SEM), transmission electron microscopy (TEM) and differential scanning calorimeter (DSC). It was found that Mo addition took significant effects on the structure and properties of AlCoCrFeNi alloy. Mo_{0.1} alloy, similar to Mo₀ alloy, was of single BCC solid solution structure. When Mo content was more than 0.1, the alloy exhibited a typical laminar eutectic structure. The alloy strength was improved obviously, but the ductility of alloy was lowered at the same time. The maximum yield strength reached 2757 MPa when Mo content was 0.5, and the maximum compressive fracture strength reached 3208 MPa when Mo content was 0.3. The strengthening effect and mechanism of Mo addition on AlCoCrFeNi alloy were discussed from different aspects.

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1. Introduction

One principal element alloy system was considered to be a conventional way to prepare alloys [1,2], because alloys with multi-principal elements were assumed to form a lot of intermetallic compounds and thus complex microstructures. However, recently, Yeh explored an entirely new class of alloy, high-entropy alloys that generally had at least five principal elements with concentrations between 5 and 35 at.% [3]. The structures of these alloys typically consisted of solid solution phase with multi-principal element. Yeh's report was completely out of anticipation [3–9]. This alloy indicated super mechanical properties, such as the yield strength and plastic strain of AlCoCrCuFeNiMo_{0.2} alloy were 1420 MPa and 3.5%, respectively [10]. Therefore, the high-entropy alloy could be an excellent material candidate that could meet different requirements of properties under different service conditions by adjusting alloy compositions in larger compositional region. In order to enrich multi-principal element alloy field, and investigate the effects of alloying elements on microstructure and properties of high-entropy alloys, the effects of Mo element on AlCoCrCuFeNi alloy had been investigated in our earlier work [10]. Cu element was rejected to grain boundary, and formed Cu-rich FCC solid solution, when Mo content was low. The segregation of Cu was inhibited

when Mo content was more than 0.2, but it resulted in the eutectic reaction of alloy. The mixing enthalpies between Cu and most other elements in alloy were large positive, which usually led to the repulsion of atoms. Therefore, the effects of Mo element on structure and properties of AlCoCrFeNi alloy that did not contain Cu element were investigated in details in this work.

2. Experimental

Alloy ingots with nominal compositions of AlCoCrFeNiMo_x (x values in molar ratio, $x=0, 0.1, 0.2, 0.3, 0.4$ and 0.5 , denoted by Mo₀, Mo_{0.1}, Mo_{0.2}, Mo_{0.3}, Mo_{0.4} and Mo_{0.5}, respectively) were prepared by arc melting a mixture of ultrasonically cleansed Al, Co, Cr, Fe, Ni and Mo with a purity of above 99.9 wt.% in a water-cooled copper hearth under Ti-gettered high-purity argon atmosphere. The chemical homogeneity was realized by repeated melting at least four times. The proper amount of ingots were then remelted under high-vacuum in a quartz tube by using an induction heating coil and injected through a nozzle with 0.5–1 mm in diameter into a copper mould with a cavity of 5 mm in diameter and 50 mm in height. The phases and microstructures were characterized using X-ray diffraction (XRD, Philips PW1050, Cu K α), scanning electron microscopy (SEM, Hitachi S3400N) accompanied with energy dispersive spectrometry and transmission electron microscopy (TEM, JEOL 2010, 200 kV). The thermal behaviors of alloys were analyzed by a high-temperature differential scanning calorimeter (DSC; Netzsch DSC 404C, Estes Park, CO) in a flowing argon atmosphere at a

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heating rate of 20 K s^{-1} . The end surfaces of compressive samples with 5 mm in diameter and 10 mm in length were polished on 2000 grit sand papers. After polishing, the samples were etched with aqua regia. Thin slices from as-cast rods were used for preparing the TEM samples, which were ground and mechanically dimpled with a GATAN precision dimple grinder as well as use argon ion milling as the final thinning process by a GATAN precision ion polishing system (PIPS). Compression test was performed on an MTS 800 machine using a strain rate $2 \times 10^{-4}\text{ s}^{-1}$.

3. Results

Fig. 1 shows the XRD patterns of as-cast AlCoCrFeNiMo_x alloys. The crystal structures of AlCoCrFeNiMo_x alloy system are characterized. Only simple BCC solid solution structure and α phase are identified. Mo_{0.1} alloy, similar to Mo₀ alloy, is of single BCC structure. The reflection peaks of α phase appear when Mo content is more than 0.1, and their intensities get stronger and stronger with the increase of Mo addition.

Fig. 2 shows the back-scattering SEM images of as-cast AlCoCrFeNiMo_x alloys. Mo₀ and Mo_{0.1} alloys show morphology of single phase, which is consistent with the XRD result. When Mo content is more than 0.1, the alloys exhibit a typical lamellar eutectic structure. The back-scattering SEM images show that the alloy consists of two distinct structures (A and B in Fig. 2(c)–(f)). The EDX results in Table 1 show that the Al content of structure A is higher than that of structure B. On the contrary, the Mo content of structure A is lower than that of structure B. As Mo content increases from

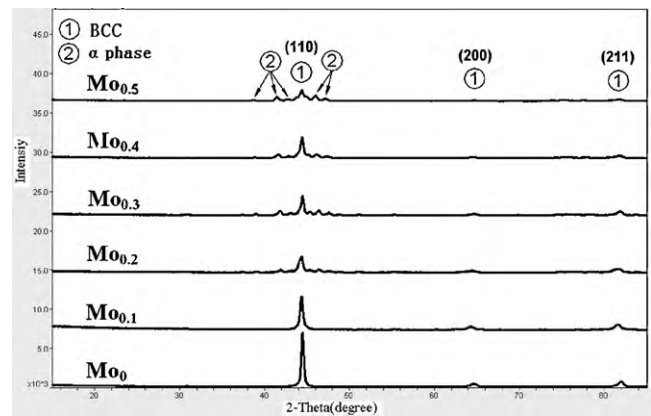


Fig. 1. XRD patterns of as-cast AlCoCrFeNiMo_x ($x = 0, 0.1, 0.2, 0.3, 0.4$ and 0.5) alloys.

0.2 to 0.5, the differences of Al and Mo content between structure A and structure B become more and more obvious.

Fig. 3 shows the TEM bright-field images and corresponding selected area electron diffraction (SAED) patterns of Mo₀, Mo_{0.1}, Mo_{0.2} and Mo_{0.5} alloys. The SAED patterns shown in Fig. 3(a) and (b) further confirm the single BCC structure of Mo₀ and Mo_{0.1} alloys. The alloy exhibits lamellar eutectic structure when Mo content is between 0.2 and 0.5. The SAED patterns shown in Fig. 3(c)–(f) confirm the formation of α phase. The interlayer distance of α phase is about 100–400 nm, which decreases as Mo content increases from 0.2 to 0.5. It is revealed by the SAED pattern shown in Fig. 3(f) that

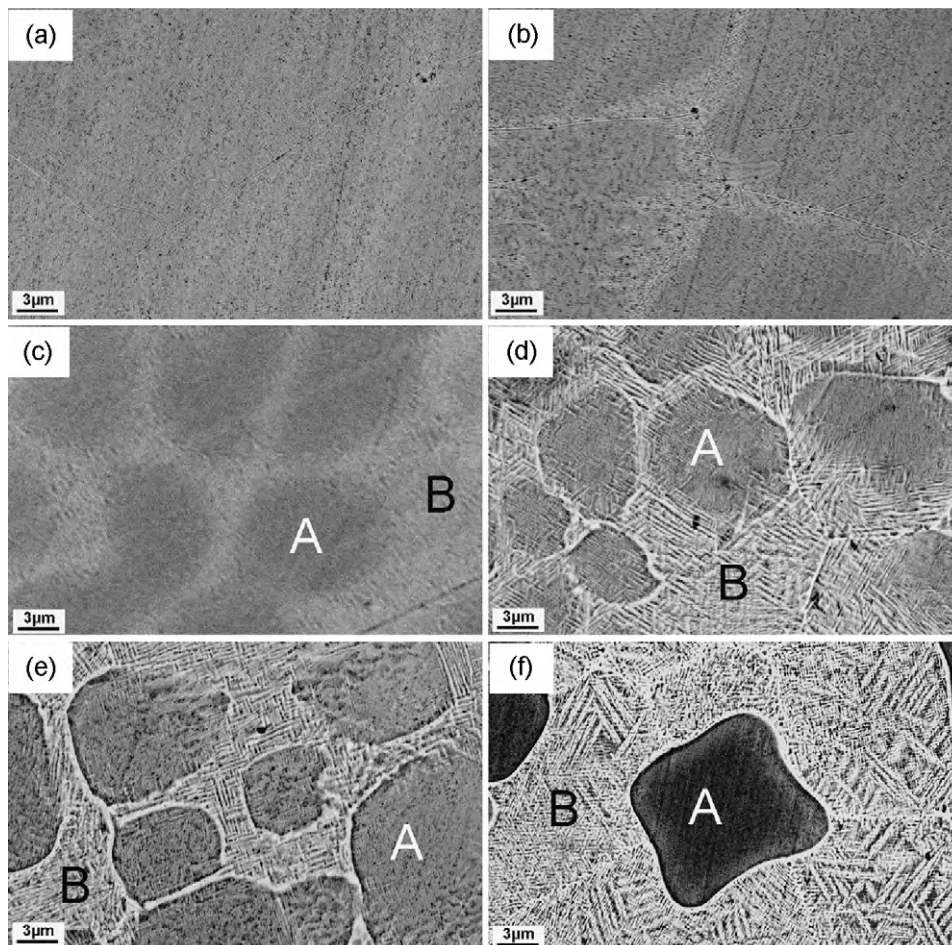


Fig. 2. Back-scattering images of as-cast AlCoCrFeNiMo_x alloys: (a) 0, (b) 0.1, (c) 0.2, (d) 0.3, (e) 0.4 and (f) 0.5.

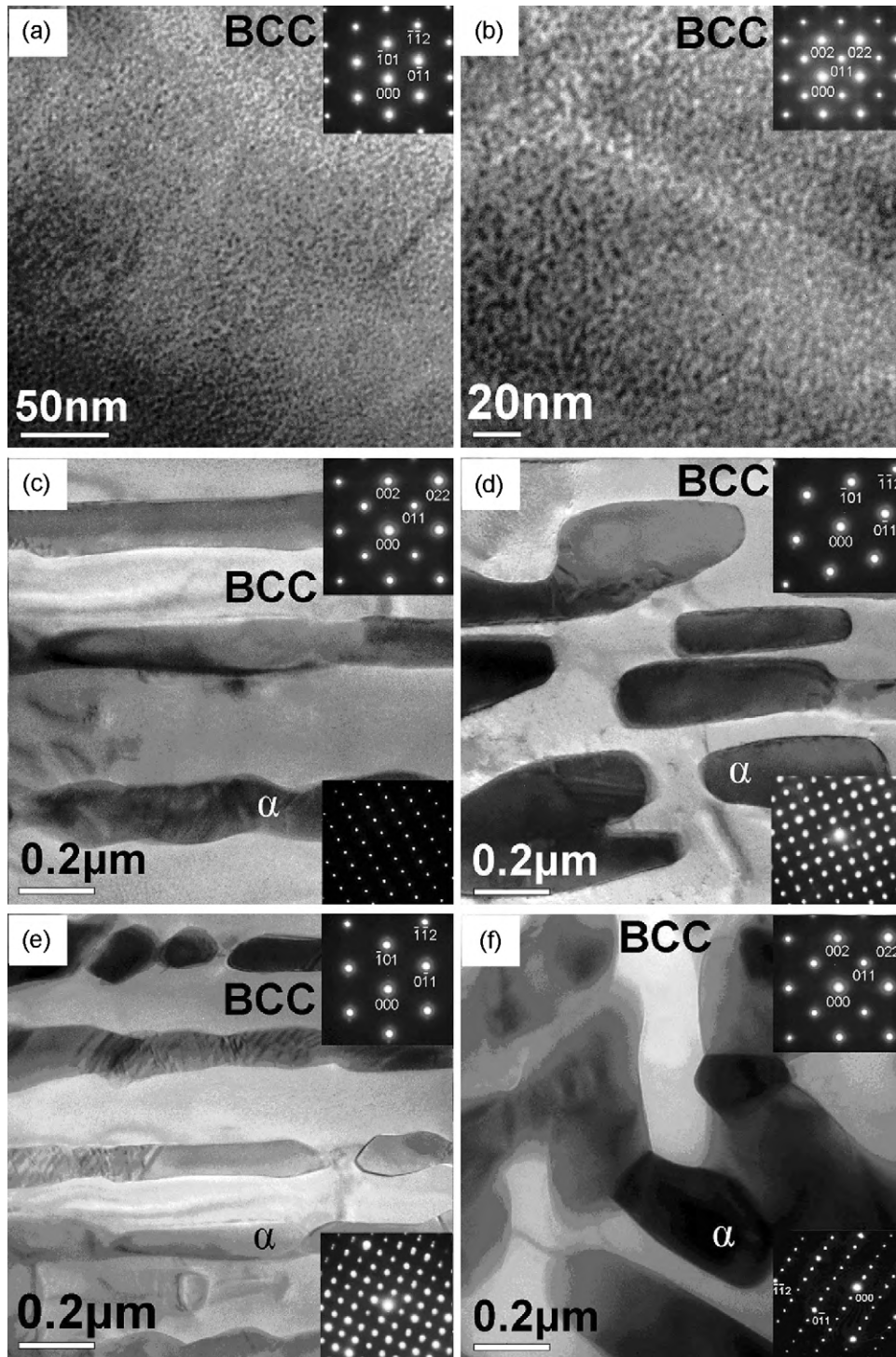


Fig. 3. TEM images (inset pictures are SAED patterns) of as-cast AlCoCrFeNiMo_x alloys: (a) 0, (b) 0.1, (c) 0.2 (structure A in Fig. 2(c)), (d) 0.2 (structure B in Fig. 2(c)), (e) 0.5 (structure A in Fig. 2(f)), (f) 0.5 (structure B in Fig. 2(f)).

there is a certain crystallographic orientation relationship between α phase and BCC phase. The EDX results in Table 1 show that the Al and Ni contents in BCC phase are higher than those in α phase. On the contrary, the Cr and Mo contents in BCC phase are lower than those in α phase. As Mo content increases from 0.2 to 0.5, the difference of Mo content between α and BCC phases become more and more obvious.

Fig. 4 shows the room-temperature compressive stress–strain curves of as-cast AlCoCrFeNiMo_x alloys. It is shown that the alloy strength is improved significantly due to the addition of Mo ele-

ment, but the ductility is weakened at the same time. Mo_{0.1} alloy, similar to Mo₀ alloy, exhibits certain ductility. The strength of Mo_{0.1} alloy is significantly higher than that of Mo₀ alloy. The yield strength σ_y , compressive fracture strength σ_{max} and plastic strain limit ε_p of the alloys are listed in Table 2. When Mo content is 0.1, the yield strength, compressive fracture strength and plastic strain limit is 1804 MPa, 2280 MPa and 9.1%, respectively. The maximum yield strength reaches 2757 MPa when Mo content is 0.5, and the maximum compressive fracture strength reaches 3208 MPa when Mo content is 0.3.

Table 1
Compositions (at.%) of structures in AlCoCrFeNiMo_x (x=0, 0.1, 0.2, 0.3, 0.4 and 0.5) alloys, compositions (at.%) of phases in AlCoCrFeNiMo_x (x=0.2 and 0.5) and atomic radii of different elements.

x	Structure or phase		Al 1.43 Å	Co 1.25 Å	Cr 1.21 Å	Fe 1.24 Å	Ni 1.24 Å	Mo 1.36 Å
0		BCC	16.7	21.1	21.2	21.9	19.1	0
0.1		BCC	19.8	19.4	21.2	19.2	18.5	2.0
0.2	A		21.0	19.5	16.5	18.1	21.5	3.4
	B		17.9	18.9	21.4	19.5	17.6	4.7
0.3	A		23.1	17.8	12.9	18.6	24.1	3.5
	B		16.1	19.3	21.0	19.5	15.5	8.6
0.4	A		24.3	19.1	16.2	16.6	19.9	4.0
	B		15.7	18.9	19.4	19.7	18.1	8.1
0.5	A		26.78	18.5	12.1	14.4	23.3	5.0
	B		14.5	18.4	20.1	19.8	15.8	11.5
0.2	A	BCC	16.9	23.2	5.8	16.9	36.2	0.9
		α	2.2	21.8	30.5	23.2	8.3	13.9
	B	BCC	12.2	23.3	8.4	17.0	37.9	1.1
0.5		α	2.6	22.4	28.5	23.1	11.3	12.1
	A	BCC	14.8	22.2	7.8	14.7	34.9	5.5
		α	2.2	17.3	21.7	17.7	8.1	33.0
	B	BCC	10.7	23.3	10.9	14.6	35.8	4.6
		α	2.4	17.5	19.6	19.8	10.0	31.1

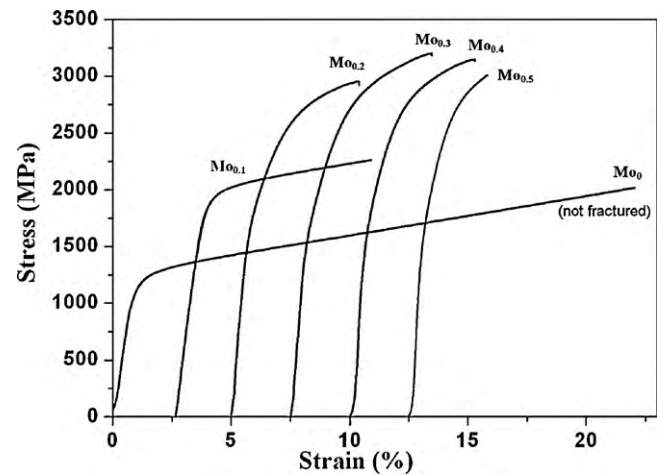


Fig. 4. Compressive stress–strain curves of as-cast AlCoCrFeNiMo_x (x=0, 0.1, 0.2, 0.3, 0.4 and 0.5) alloys.

Table 2
Mechanical properties of as-cast AlCoCrFeNiMo_x (x=0, 0.1, 0.2, 0.3, 0.4 and 0.5) alloys.

x	Yield stress σ _y (MPa)	Compressive strength σ _{max} (MPa)	Plastic strain ε _p (%)
0	1051	–	–
0.1	1804	2280	9.1
0.2	2456	2953	3.4
0.3	2649	3208	3.3
0.4	2670	3161	3.0
0.5	2757	3036	2.5

Fig. 5 shows the DSC results of as-cast AlCoCrFeNiMo_x alloys. The curves of Mo₀ and Mo_{0.1} alloys show an obvious endothermic peak at around 620 °C, which disappears when Mo content is more than 0.1. The curves of all six alloys show an endothermic peak at high-temperature, of which the intensities for alloys with Mo addition are clearly larger than that for Mo₀ alloy. It can be seen

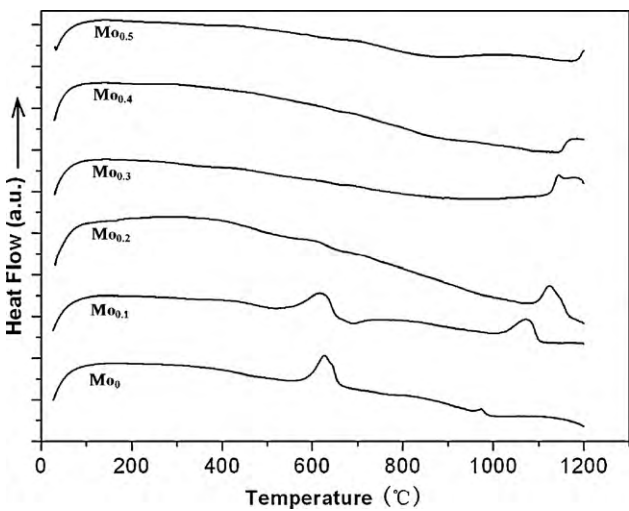


Fig. 5. DSC curves of as-cast AlCoCrFeNiMo_x (x=0, 0.1, 0.2, 0.3, 0.4 and 0.5) alloys.

that the endothermic peaks at high-temperature shift rightwards obviously as Mo content increases from 0 to 0.5.

4. Discussion

The effects of Mo element on AlCoCrFeNi alloy are similar to those on AlCoCrCuFeNi alloy [10]. There is FCC solid solution phase in AlCoCrCuFeNiMo_y (y=0, 0.2, 0.4, 0.6, 0.8 and 1.0 molar ratio) alloy system when Mo content is low, but there is no FCC solid solution phase in AlCoCrFeNiMo_x alloy system. The FCC solid solution phase, which is rich in Cu element, disappears when Mo content is more than 0.2 since the Mo element inhibits the Cu segregation. The excessive addition of Mo element on both of AlCoCrCuFeNi and AlCoCrFeNi alloys will result in the formation of eutectic structure, α phase, which is produced by eutectic reaction and rich in Mo element, is observed in both of AlCoCrFeNiMo_x and AlCoCrCuFeNiMo_y alloy system when Mo content is high. In AlCoCrCuFeNiMo_y alloy

Table 3
Lattice constants of BCC phase in AlCoCrFeNiMo_x (x=0, 0.1, 0.2, 0.3, 0.4 and 0.5) alloys.

x	0	0.1	0.2	0.3	0.4	0.5
Lattice constant (Å)	2.878	2.886	2.885	2.883	2.880	2.878

system, the maximum yield strength reaches 1920 MPa when y is 0.8, and the compressive fracture strength reaches 2820 MPa when y is 0.6. In AlCoCrFeNiMo $_x$ alloy system, the maximum yield strength reaches 2757 MPa when x is 0.5, and the maximum compressive fracture strength reaches 3208 MPa when x is 0.3.

According to the Gibbs phase rule, $F = C - P + 1$ (F , degree of freedom; C , number of components, and P , number of phase), the maximum number of phases in equilibrium in a C component system at constant pressure is $P = C + 1$. However, in this research, only BCC and α phases are identified, and the total number of phases is well below the maximum equilibrium number allowed by Gibbs phase rule. This is attributed to the effect of high-entropy of mixing [8]. Among the six elements of alloy, the atomic radius of Al is the biggest one, and that of Mo comes next. The mixing enthalpies between Al and Co, Cr, Fe, Ni and Mo are -10 , -10 , -11 , -22 and -5 kJ/mol, respectively [11]. The mixing enthalpy between Al and Mo is larger than those between Al and other elements. This is the reason why Mo segregates to structure A and Al segregates to structure B when Mo content is between 0.2 and 0.5.

The lattice constants of BCC phases for all six alloys are calculated based on the XRD results, as shown in Table 3. It is found that the lattice constants of BCC phase for Mo $_{0.1}$ alloy are larger than those for other alloys. The improved mechanical properties of Mo $_{0.1}$ alloy are attributed to solid solution strengthening mechanism of Mo atoms. The atomic radii of different elements are listed in Table 1 [12]. As the Mo atom occupies the lattice sites, the lattice distortion energy will increase significantly and the effect of solid solution strengthening is enhanced, thus the alloy strength greatly increases with the decrease of ductility.

When Mo content is more than 0.1, the solid solubility of Mo element in alloy reaches its limit. The alloy appears with eutectic reaction. The formation of α phase, of which the Mo content is much higher than that of BCC phase, can release the lattice stress of BCC solid solution structure effectively. This is the reason why the lattice constants of BCC solid solution phase decrease as Mo content increases from 0.1 to 0.5. It is revealed by XRD and SAED pattern that the structure of α phase is complicated. The number of slip systems in complex structure is lower than that of bcc structure, so the ductility of α phase is smaller than that of BCC phase. This is the reason why the ductility of alloy is lowered largely. The interlayer distance of α phase is about 100–400 nm, which decreases as Mo content increases. It is revealed by the SAED pattern shown in Fig. 3(f) that there is a certain crystallographic orientation relationship between α and BCC phase. This means that there is a strong mutual reinforcing effect between α and BCC phase. Moreover, the formation of laminar structure will result in great increase of phase interface, which can effectively enhance the alloy strength. This

is the reason why the alloy strength increases significantly. The low-temperature phase transformation of alloy is inhibited, and the temperature of high-temperature phase transformation of alloy is raised due to the addition of Mo element. It can be concluded from the DSC results that the addition of Mo element is beneficial to the high-temperature properties of alloy.

5. Conclusions

In summary, the alloy series of AlCoCrFeNiMo $_x$ have relatively simple microstructure and promising properties. Mo $_0$ and Mo $_{0.1}$ alloys show morphology of single phase. α phase appears when the Mo content is more than 0.1, and its amount increases as the Mo content increases. The yield strength of alloy increases from 1051 to 2757 MPa and the compressive fracture strength of alloy increases from 2280 to 3036 MPa with the increase of Mo addition, among which the Mo $_{0.3}$ alloy exhibits highest strength up to 3208 MPa, that should be attributed to solid solution of Mo element and precipitation strengthening of α phase. It can be predicted that the alloy has promising high-temperature properties.

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